Linked Data in Libraries

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SLIS 5010: Libraries, Culture, and Society

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December 8, 2023

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Information is dynamic: Our world and what we know about it is constantly growing and changing. How then do we classify it in a meaningful way that is relevant for any length of time? Current practice has each institution with its own catalog and its own records. They may or may not pull authority records from a collective source and those authorities may or may not be implemented in the way their creators intended. Their creators might have perspectives that are limited, outdated, or offensive today. Even with the best of intentions, staff may not be able to fix every problem that arises from having a catalog that essentially exists in a vacuum, whether through lack of time, skill, or awareness. These inherent issues lead to barriers that prevent a library's community from having full access to the knowledge the library holds. Fortunately (or unfortunately) our modern world is globally connected. We have the ability to develop new resources to overcome these barriers and solve these problems. Linked data is one of these solutions.

What is Linked Data?

In 1989, Tim Berner-Lee created an information space called World Wide Web. It began as a collection of documents, or information intended for humans and expressed in everyday language that is easy for humans to comprehend, and that's still how it mostly is today. Following the development of the internet, the he proposed the Semantic Web, which is an information space composed of a collection of data, established to express information through machine-readable relationships. Instead of information being informally linked, data is semantically interlinked, creating a web that allows information within a page to be connected to other information, not just pages linked to pages (DeWeese & Segal, 2014) (TED, 2009).

Linked data is a framework for organizing data in a way that includes relationships between objects and emphasizes the synergistic benefits of large, interconnected datasets that are freely

available. The results is the Semantic Web. Using linked data, a single piece of information in one database can be associated with the same piece of information in another database, using a relationship that expresses that the data are equal, thus connecting the databases to each other (DeWeese & Segal, 2014). This is the beauty and ingenuity of linked data: a single connection can reveal a wealth of information that would have otherwise never been realized.

Linked data does not require any specific methods (Alemu et al., 2014), however, librarians generally encounter a particular set of standards when working with linked data, some of which are discussed below.

URIs

Uniform Resource Indicators (URIs), a foundational concept for linked data, are defined as "a compact string of characters for identifying an abstract or physical resource" (IETF, 2005 via DeWeese & Segal, 2014). Two common types of URIs that many people are already familiar with are the URL, which is a unique identifier for a webpage, and the ISBN, which is a unique identifier for a book edition. The significance of a URI is that a document can contain many other URIs that identify specific elements of the document. It is not necessary to only refer to the document itself.

RDF

One common model for the basis for linking data is Resource Description Framework (RDF), which uses a "triple" to describe properties of and relationships between data. Triples are built (unsurprisingly) with three parts: a subject, an object, and a predicate. A subject is an information resource; an object is either an information resource or a data value; a predicate is a description of the relationship between the subject and the object. The subject and predicate are always represented by a URI. An object is either represented by a URI or is simply a data value. Thus, the URI for a specific

edition of a book can be associated to the URIs of an author, a publisher, a title, a publication year, and so on, using URIs that represent each relationship (DeWeese & Segal, 2014). The benefits of RDF include a consistent framework and standard syntax that allows metadata to be shared, exchanged, and queried more easily and precisely (Eye on Tech, 2022).

RDF has been expanded upon in various ways. RDF Schema (RDFS) adds hierarchical relationships, annotations, and restrictions to create more complex descriptions of information resources. The Library of Congress is using these kinds of relationships in their proposed replacement for MARC, called BIBFRAME. The next level of expansion is Web Ontology Language (OWL), which provides the capability to describe more complex data models, including the especially-useful descriptor for equivalence. Further expansions that are specific to a particular domain of knowledge are known as vocabularies or ontologies (DeWeese & Segal, 2014).

SPARQL

SPARQL Protocol and RDF Query Language (SPARQL) is a common query language used to search and manipulate RDF data, although it often runs slowly and is not well-supported by all open data services (Tillman, 2019). Tools are being created and improved to translate natural language queries into SPARQL, “bridg[ing] the gap between users and ontologies” (Ochieng, 2021).

Five Star Linked Open Data

In order for information to be formally linked, it must be uniquely identified, freely available, standardized, and linked in a way that allows discovery of more information (DeWeese & Segal, 2014). Berners-Lee has developed a 5-star rating system for linked open data. All levels require data to be open, or freely available at no charge through an open license such as Creative Commons CC-BY, and

assume that the next highest level meets the requirements of the levels below it (2009). The levels for quality of open data as described by Berners-Lee are as follows:

- Available on the web in any format
- Available as machine-readable structure data
- Available in a non-proprietary format (e.g. .csv vs .xml)
- Use open standards such as RDF and SPARQL to identify things
- Link your data to other people's data to provide context

Linked Data in Library Catalogs

One area where linked data is being studied and implemented is in library catalogs. The current standard commonly used in the US is MAchine Readable Cataloging (MARC). It was designed in the 1970s, before the internet, and the intention was to computerize card catalogs, not change how they were structured (Arvam, 1975), and thus doesn't support linked data well (Schreur, 2018).

In contradiction to the ALA's core value of supporting diverse communities and even despite intentions to reflect collection diversity, issues of bias and othering persist in library catalogs (Yon & Willey, 2022). Popular controlled vocabularies like the Library of Congress Subject Headings (LCSH) are helpful to libraries and other institutions in creating standards of terminology and information access points, but these vocabularies are often slow to change and exhibit a point of view that marginalizes diverse and minority communities.

A lack of the specialized knowledge of controlled vocabularies, as well as their use of potentially offensive and harmful language, can create barriers to the effectiveness of using the vocabularies to find resources (Hardesty & Nolan, 2021). Unchanging or permanent metadata can also

exclude creators whose identities have changed over time but may no longer want to be recognized by a previous descriptor, such as a specific gender or name (Yon & Willey, 2022).

Allowing for the integration of socially-constructed or user-created metadata can reduce patron barriers (Alemu et al., 2014) and organizations like Homosaurus are working to create their own controlled vocabularies to support this (Hardesty & Nolan, 2021) (Fischer, 2023).

Linked data presents an opportunity for information science but there are also drawbacks that need to be addressed and barriers to implementation at the library level before the system can be fully realized. Despite efforts, development of vocabularies and other frameworks for linked data are often short-term, experimental, and project-based. Problems arise when a database is not maintained or even entirely disappears, leaving gaps in the other databases that relied upon it, weakening the usefulness of the entire interconnected system. The linked data system in its current form lacks the robustness to be trustworthy enough for most institutions to adopt, though paradoxically adoption of the technology could create the kind of robustness that libraries are looking for (Salo & Tillman, 2023). Other barriers are how to handle the metadata that already exists in the catalog (Schreur, 2018) and lack of best practices for doing so, as well as the customary lack of resources that comes with the nature of librarianship (LaPolla, 2013).

As described above, an open metadata approach to linked data is ideal, however this requires not only the assumption that any description is valid, but also the encouragement of, although not necessarily the agreement with, multiple interpretations of metadata (Alemu et al., 2014) (Brown, 2022). While this parallels the kind of thinking seen in library ethical standards, Tillman points out, “we cannot expect the technology to be any less vulnerable to exploitation than others” (2019) and so we must balance openness with harm reduction.

Linked data does not currently convey whether or not the information represented is universal or contemporary. Validation of information and annotation of information as a viewpoint is necessary to

make sure people have access to correct information. For example, annotations could provide context like the geographic location where a term is primarily used or the community that has a particular practice. This would also allow for searching within a particular viewpoint (Choi, 2023). Other concerns include implementation of standards that have their basis in colonial practices, the creation of problematic or incorrect relationships, and the provenance of collection and data (Allison-Cassin, 2022).

Conclusions

While linked data is still a developing technology, and there are obstacles to be overcome before it is fully functional, in my opinion, the benefits of achieving a robust connected linked data system that has been thoughtfully and ethically managed outweigh the costs of implementation and transition. As much as libraries have been working to support diversity and inclusion efforts, it makes sense that these efforts would not only include the physical spaces and materials found in a library, but also the metadata that underlies information discovery.

In order to truly eradicate the systems of oppression and overturn the legacy of colonization, we need to create something entirely new that doesn't rely on the foundations that were built in the past. Even if the authority records maintained by the Library of Congress were able to be changed quickly and efficiently, they leave no room for differences in opinion or interpretation, and no room for the messy reality that our world is. How then could they hope to represent it? Linked data might not be *the* solution, but it is *a* solution, one which librarians should turn their considerable problem-solving skills towards, at least until something better comes along.

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